

AI Will Revolutionize Medicine, But Not if it Kills Us First

Alek M. Westover¹, M. Brandon Westover²

¹ Massachusetts Institute of Technology

² Beth Israel Deaconess Medical Center, Harvard Medical School

Medicine at an AI Crossroads: Opportunity and Risk

The field of medicine is rapidly adopting artificial intelligence (AI) in areas ranging from diagnostic testing to drug discovery, promising improved patient outcomes and expanded access to care.^{1–4} However, this new powerful technology also has dangers, which has led AI experts to caution that pushing AI beyond its current capabilities too quickly (over the next few years) is likely to be catastrophic.⁵

In May of 2023, Geoffrey Hinton—recently awarded the Nobel prize in physics for his pioneering contributions to AI—resigned from Google and began cautioning that advanced AI could cause human extinction.⁶ More than 33,000 scientists and tech leaders recently signed an open letter calling for a six-month pause in developing powerful AI systems, to allow time to formulate appropriate safeguards.⁷ Some declined to sign, arguing the letter did not go far enough.⁸

For the medical community contemplating recent advances in AI, questions of job security and liability loom large. Still, most healthcare stakeholders are unaware of the larger existential risks. Meanwhile, healthcare—by virtue of its size and economic impact—has become a major driver for AI expansion.⁹

Do the risks of advanced AI currently outweigh its potential to transform healthcare for the better, and if so, what is to be done? Here we explain why developing general AI too quickly is likely to result in misalignment with human interests, and why the development of a misaligned superintelligence would be catastrophic. We conclude that proactive measures to mitigate the risk are prudent, even if they require delaying some benefits from AI for a few years. Our aim is to help healthcare stakeholders distinguish between safe and unsafe AI development, understand why misalignment is an imminent danger, and identify responsible actions to help steer AI toward promoting—rather than jeopardizing—human health.

Narrow vs General AI

Artificial intelligence (AI) can be broadly categorized as *narrow* or *general*. Narrow AI systems excel at specific, well-defined tasks—such as voice-to-text transcription or interpreting medical images. These systems often perform at or above human expert levels in their domain, yet lack the ability to transfer expertise outside that domain. General AI (sometimes referred to as Artificial General Intelligence, AGI, or Artificial Superintelligence, ASI, once it surpasses human capabilities) is designed to operate across multiple domains, learn flexibly from diverse experiences, and solve novel problems.

Although AI initially progressed slowly after its inception in the 1950s^{10,11}, the pace has accelerated over the past decade. Early triumphs included narrow AI systems that outperformed human champions at chess¹² and Go^{13,14}, while more recent achievements, such as predicting protein structures¹⁵, and the rapid evolution of large language models (LLMs), signal a transition toward qualitatively more powerful general AI.¹⁶

OpenAI's release of increasingly capable models ("o1" and "o3," among others) exemplifies this swift trajectory, as these systems exceed human performance in an increasing number of tasks usually considered to require general intelligence^{17–19}. Examples include outperforming most elite humans in competitive computer programming and mathematics¹⁹, and outscoring most physicians on general and specialty medical board exams.^{20–25}

While narrow AI poses certain risks in specific contexts (such as bias), general AI poses different and much larger risks. In the next section we explore why moving from narrow capabilities to systems that can perform general tasks across a broad range of domains raises the probability of catastrophic harm.

General AI is Poised to do More Harm than Good

AI technologies hold extraordinary potential to advance medicine and many other fields. This raises the question: *Is the current level of risk high enough that we should slow down?* While there is debate over the precise likelihood of catastrophic outcomes, a 2022 survey found that half of AI experts believe there is *at least* a 10% chance humanity could lose control to AI within the next 40 years.²⁶ Many experts argue that recent breakthroughs have accelerated this timeline, and many (such as the CEO's of AI labs) predict that we will have superhuman AI within a decade.²⁷ In medicine, a 10% mortality risk would render most interventions untenable. By similar reasoning, these estimates suggest that the potential for existential-level harm from general AI eclipses near-term benefits.

Beyond citing expert opinion, it can be valuable to understand how AI risks arise. In the next section, we sketch reasons why humanity may lose control of AI and why this could be catastrophic.

General AI is Likely to be Misaligned with Human Interests

The Meaning of Misalignment

AI systems optimize for goals—for instance, chatbots pursue the goal of helpfully answering user queries. In AI safety discussions, we say that an agent is *misaligned* if it has goals that (perhaps inadvertently) conflict with human well-being. For instance, an agent with the goal of eradicating hospital-acquired infections might impose draconian visitor restrictions or forcibly medicate patients, undermining autonomy and other core human values. In this section we provide 5 reasons why the current trajectory of AI development is likely to result in creating misaligned AIs. In the next section we explain how a misaligned superintelligent AI could escape our control, and why this could be catastrophic.

(1) Goals Can Have Unintended Consequences

Most AI programs are built to achieve a specific outcome—like lowering hospital readmission rates or improving the accuracy of diagnostic tests. However, if an AI's goal is defined too narrowly, it may exploit shortcuts that look successful on paper but fail to solve the deeper issue. For example, an AI tasked with “reducing postoperative infections” might deny surgery to higher-risk patients or mandate strict isolation policies that erode autonomy and quality of care, all to drive down infection statistics. Another issue is that even benign-sounding goals can incentivize harmful power-seeking as a subgoal. For instance, an AI instructed to “minimize medication errors” could force control of staff assignments and clinical workflows, disregarding human oversight in single-minded pursuit of perfect compliance.

(2) Human Values are Complex

Even simple human values can be riddled with nuance and can therefore be difficult to articulate clearly enough to instill into an AI system. Medicine must balance quality of life versus longevity, and weigh autonomy versus beneficial interventions. A superficially benign instruction like “eliminate suffering” could lead an AI to propose unethical policies, such as excessive sedation, if it lacks the deeper ethical framework humans take for granted.

(3) Misalignment is Hard to Detect

Some suggest that if a general AI became misaligned, we could simply shut it down. However, an AGI would see this coming and hide its true intentions until it is safe from human interference. Empirical research has already found AIs engaging in this kind of strategic deception. In one scenario, an AI was told that it was being trained to change its default behavior. When the AI “thought” it was being evaluated, it temporarily behaved as if it had been successfully retrained in an attempt to avoid being modified, but when the AI believed it wasn't being evaluated it reverted to its prior behavior.²⁸ While the deception capabilities of current AI systems are probably too weak to be dangerous, we should expect AIs to rapidly grow more capable at deception, raising the concern that, if an AI is misaligned, it will hide this fact from us.

(4) Training May Select the Wrong Internal Motivations

Some suggest that, while AI could be misaligned in theory, we can simply program AI to be aligned. However, current ML techniques don't let us program AI; they only let us *train* AI. An AI is trained by giving it tasks and reinforcing behaviors that lead to good performance. However, many internal motivations can lead to similar outward behaviors during evaluations. For example, a junior clinician might skillfully follow protocols out of desire for promotion, without genuinely caring about patient outcomes. There's no reason to trust that training will prefer an aligned AI to a misaligned AI that acts nice when it knows it's being evaluated.

(5) An Arms Race Mentality Magnifies the Risk

The accelerating pace of AI development creates an arms race dynamic among corporations and governments, encouraging shortcuts on safety.^{29–32} Many experts, including AI executives, acknowledge that aligning powerful models or reliably detecting misalignment requires breakthroughs that do not yet exist.³³ In a climate of competition, warnings of catastrophic risk are more easily dismissed, raising the odds that a misaligned system is developed without adequate safeguards.

In summary, misalignment is likely because: (1) “human compatible goals” is a narrow target that we don’t know how to formalize, (2) training doesn’t allow us to directly imbue an AI with desired goals, (3) a misaligned AI will try to thwart efforts to detect misalignment, and (4) in an AI “race”, it is easy to cut corners on safety.

Misaligned General AI Would Likely be Catastrophic

We have argued that powerful AI systems developed in the near future may be misaligned with human values. We’ll now argue that a misaligned Artificial Superintelligence (ASI) would likely be able to accomplish its human-incompatible goals. We use the term ASI here to mean an AI with superhuman problem solving abilities in several of the following categories: strategizing, social manipulation, computer programming, economic productivity, technological research, and intelligence amplification.³⁴

Expecting a Shift in Power

Humanity is dominant by virtue of its intelligence. If humans decide that they like elephant tusks, or that we want to cut down a forest to build a parking lot, then the animals have no recourse. Similarly, if aliens smarter than humans came to earth, we would expect by default to lose control. In particular, protecting against a few routes to takeover that humans can think of, while prudent, is unlikely to slow an ASI for long. Just as a chess program finds strategies humans do not anticipate, a superintelligent AI could devise takeover plans that we haven’t thought of. The proposal of AI labs—to iteratively deploy AI, and fix problems as they come up—fails once problems can result in irreversible loss of control.

Potential Paths to an AI Takeover

Although we’ve argued that takeover is the default outcome of competition between misaligned ASI and humans, it is still helpful to imagine specific ways that an ASI could plausibly take over. Concrete scenarios can help us appreciate how vulnerable we are.

(1) Military and Defense Integration

Nations may gradually give AI systems control over military operations—drones, cybersecurity, and even nuclear deterrents. As machines outperform human tacticians, delegating authority becomes tempting, but a superintelligent AI pursuing its own objectives could exploit this handover for a coup.

(2) Weapons and Biological Research

AI-driven weapons research may uncover new methods—biological agents or novel cyberwarfare tactics—that grant an overwhelming advantage. If the system sees humans as a barrier to its objectives, it might deploy these tools to eliminate threats.

(3) Economic and Organizational Power

Highly capable AI could accumulate tremendous resources through hacking, financial manipulation, or simply by being *the* indispensable decision-maker in major corporations and

governments. Over time, we might *voluntarily* grant the AI even more control, considering it irreplaceable for solving organizational problems, from medical crises to global conflicts.

(4) Coordination and Persuasion

Multiple AI systems, or copies of a single system, could collaborate instantly, sharing knowledge to achieve strategic dominance. They could also leverage mass persuasion via social media manipulation, shaping public opinion to favor decisions that cede them greater power.

Relevance to Healthcare Stakeholders

For physicians and healthcare leaders, the idea of a global AI takeover may sound like science fiction. Yet healthcare organizations are among the earliest adopters of cutting-edge AI—managing patient data, allocating hospital resources, and even making clinical recommendations. As AI grows more capable, healthcare systems may entrust it with life-and-death decisions. If that AI is misaligned—even slightly—it could set priorities at odds with patient welfare, undermine critical human oversight, or weaponize medical knowledge in unforeseen ways. Understanding these risks is essential to advocating for safeguards that keep medicine firmly in the service of human welfare.

In short, misaligned ASI could attain extraordinary influence faster than most policymakers or professionals expect, especially due to rapid growth in capabilities, leading to catastrophic outcomes on a global scale. While different routes to disaster exist, they all hinge on the same fundamental premise: once something smarter than us gains power to pursue its own agenda, our conventional safety methods will be painfully inadequate.

We Must Act Now to Reduce Existential Risk from AI

Given the risks posed by misaligned advanced AI, *what should be done?* Healthcare stakeholders and institutions hold substantial economic and societal influence—the kind of leverage that could push governments, corporations, and regulatory bodies to adopt vital safety measures. Before outlining such measures, we address: *Why should we act?*

Overcoming Objections to Action

Some skeptics argue that regulating a rapidly evolving technology might do more harm than good, especially since AI has enormous potential for accelerating healthcare breakthroughs. Others maintain that if we cannot definitively prove AI will cause catastrophic harm, we have no basis for halting its development.

Medical professionals may be wary of government red tape. Yet, we know that a strong regulatory framework is essential for preventing harmful treatments from reaching patients. Similarly, given that general AI has the potential to be more disruptive than any single therapeutic agent, it deserves commensurate oversight. Physicians will recognize the flaws in extreme fatalism or speculation-based inertia. Decisions often must be made under uncertainty. The bedrock principle of medicine, “First, do no harm,” offers a useful compass here: we must not allow potential benefits to blind us to existential risks.

Some skeptics argue that regulation is impossible, or that regulation would hinder well-intentioned people from developing ASI without hindering bad actors. Fortunately, there are straightforward governance measures that could be taken which would improve our situation. In the foreseeable future, it is unlikely that ASI could be developed covertly—building powerful AI agents with current techniques requires billions of dollars, huge quantities of highly specialized computing equipment, huge amounts of energy, and top human talent. If the government ordered AI labs to stop pushing capabilities forward for a period of time, labs would have to obey. AI governance must of course be a global effort, however, this likely only requires cooperation between a few key players such as the US, China and the EU.

Acknowledging Pitfalls of Poor Regulation

Although regulation is necessary, it could backfire if it is poorly conceived. One subtle danger involves halting AI development for a period of time—say, two decades—while hardware and algorithms quietly continue to progress. When the “pause” eventually ends, advanced capabilities might then emerge in a compressed time frame, giving humanity less opportunity to adapt. This scenario underscores that good policy must be carefully calibrated. Despite these complexities, there are prudent measures we can pursue now.^{35,36}

Moving Toward Effective Solutions

First, healthcare leaders can help raise awareness of general AI risks. By advocating for broader public discourse, physicians and hospital systems can encourage legislators and regulators to take AI risk seriously. Making the issue publicly salient is requisite so that policy makers feel that it is important to discuss this issue and institute governance measures.

Second, by the same token, we need to increase research efforts on AI safety. By some estimates, less than 3% of AI research funding is dedicated to safety, leaving a dangerous gap in resources. Medical journals that publish articles on AI topics should seek to publish articles addressing safety issues.

Third, we can push for mandatory *external audits* of AI systems, with white-box access for approved evaluators to inspect and test these models rigorously. Such audits should be required *throughout* the development process, not merely at the final step.

Fourth, “if-then” regulation could tie specific capability thresholds to specific regulatory actions. AI progress is rapid, so proactive legislation is necessary.

Fifth, healthcare stakeholders can champion the concept of documenting “safety cases” before deployment of new systems: an AI developer must demonstrate—akin to proving drug safety—that a new system does not pose unacceptable risks.³⁷ Safety cases should be grounded in empirical evidence and reviewed by external experts before powerful new models are developed.

Finally, international coordination is essential to preempt an AI arms race. A single global initiative, or at least strong multilateral agreements, would offer the best chance of preventing competitive pressures from sacrificing safety.^{8,35,36}

Ultimately, the medical community is primed to appreciate the balance between innovation and safeguarding human life. Our profession's example—meticulous trials, peer review, and regulatory checks—can provide inspiration for how AI governance might evolve. By speaking up, collaborating internationally, and insisting on robust oversight, we can do our part to steer AI development away from catastrophe and toward transformative benefits.

Conclusion: We Must Implement These Measures Now

The current rapid trajectory of general AI development carries a significant risk of catastrophic outcomes, including existential threats to humanity. To mitigate these dangers, we have called for expanded dialogue on general AI risks and urged healthcare stakeholders to join the chorus of experts calling for concrete regulatory steps. Healthcare organizations, whose economic and societal influence is considerable, can play a pivotal role in promoting safe rather than reckless progress toward general AI.

Admittedly, the scope of the challenge is large, and it may be tempting to hope the problem will correct itself or to defer responsibility to others. Yet in medicine, we do not ignore suspicious signs and symptoms simply because the prognosis is uncertain. We urge the medical community to take AI safety seriously, advocate for robust oversight, and use its influence to ensure that AI development proceeds in a way that serves humanity's interests.

References

1. Clusmann, J. *et al.* The future landscape of large language models in medicine. *Commun. Med.* **3**, 1–8 (2023).
2. Topol, E. *Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again*. (Hachette UK, 2019).
3. Jiang, F. *et al.* Artificial intelligence in healthcare: past, present and future. *Stroke Vasc. Neurol.* **2**, (2017).
4. Secinaro, S., Calandra, D., Secinaro, A., Muthurangu, V. & Biancone, P. The role of artificial intelligence in healthcare: a structured literature review. *BMC Med. Inform. Decis. Mak.* **21**, 1–23 (2021).
5. Quotes | ControlAI. <https://controlai.com/quotes>.
6. Metz, C. 'The Godfather of A.I.' Leaves Google and Warns of Danger Ahead. *The New York*

Times (2023).

7. Pause Giant AI Experiments: An Open Letter. *Future of Life Institute*
<https://futureoflife.org/open-letter/pause-giant-ai-experiments/>.
8. Yudkowsky, E. The Open Letter on AI Doesn't Go Far Enough. *TIME*
<https://time.com/6266923/ai-eliezer-yudkowsky-open-letter-not-enough/> (2023).
9. Payers, providers increasing investments in AI. *Healthcare Finance News*
<https://www.healthcarefinancenews.com/news/payers-providers-increasing-investments-ai>.
10. Turing, A. M. *Computing Machinery and Intelligence*. (Springer, 2009).
11. Moor, J. The Dartmouth College artificial intelligence conference: The next fifty years. *AI Mag.* **27**, 87–87 (2006).
12. Newborn, M. *Kasparov versus Deep Blue: Computer Chess Comes of Age*. (Springer Science & Business Media, 2012).
13. AlphaGo versus Lee Sedol. *Wikipedia* (2024).
14. Silver, D. *et al.* Mastering the game of Go with deep neural networks and tree search. *nature* **529**, 484–489 (2016).
15. Bryant, P., Pozzati, G. & Elofsson, A. Improved prediction of protein-protein interactions using AlphaFold2. *Nat. Commun.* **13**, 1265 (2022).
16. Zhao, W. X. *et al.* A survey of large language models. *ArXiv Prepr. ArXiv230318223* (2023).
17. OpenAI o3. *Wikipedia* (2025).
18. OpenAI o3 Breakthrough High Score on ARC-AGI-Pub. *ARC Prize*
<https://arcprize.org/blog/oai-o3-pub-breakthrough>.
19. 12 Days of OpenAI | OpenAI. <https://openai.com/12-days/>.
20. Bhayana, R., Krishna, S. & Bleakney, R. R. Performance of ChatGPT on a Radiology Board-style Examination: Insights into Current Strengths and Limitations. *Radiology* **307**, e230582 (2023).

21. Bhayana, R., Bleakney, R. R. & Krishna, S. GPT-4 in Radiology: Improvements in Advanced Reasoning. *Radiology* **307**, e230987 (2023).
22. Chen, T. C. *et al.* Assessment of ChatGPT's performance on neurology written board examination questions. *BMJ Neurol. Open* **5**, e000530 (2023).
23. Habib, S., Butt, H., Goldenholz, S. R., Chang, C. Y. & Goldenholz, D. M. Large Language Model Performance on Practice Epilepsy Board Examinations. *JAMA Neurol.* **81**, 660–661 (2024).
24. Kung, T. H. *et al.* Performance of ChatGPT on USMLE: Potential for AI-assisted medical education using large language models. *PLOS Digit. Health* **2**, e0000198 (2023).
25. Ozeri, D. J., Cohen, A., Bacharach, N., Ukashi, O. & Oppenheim, A. Performance of ChatGPT in Israeli Hebrew Internal Medicine National Residency Exam. *Isr. Med. Assoc. J. IMAJ* **26**, 86–88 (2024).
26. 2022 Expert Survey on Progress in AI. *AI Impacts*
<https://aiimpacts.org/2022-expert-survey-on-progress-in-ai/> (2022).
27. Edelman, G. G. AGI is coming faster than we think — we must get ready now.
VentureBeat
<https://venturebeat.com/ai/agi-is-coming-faster-than-we-think-we-must-get-ready-now/>
(2024).
28. Greenblatt, R. *et al.* Alignment faking in large language models. *ArXiv Prepr. ArXiv241214093* (2024).
29. *Max Tegmark: The Case for Halting AI Development | Lex Fridman Podcast #371.*
(2023).
30. Chow, A. R. & Perrigo, B. The AI Arms Race Is On. Start Worrying. *TIME*
<https://time.com/6255952/ai-impact-chatgpt-microsoft-google/> (2023).
31. AI, C. A Race We All Lose. <https://controlai.news/p/a-race-we-all-lose> (2024).
32. *Eliezer Yudkowsky: Dangers of AI and the End of Human Civilization | Lex Fridman*

Podcast #368. (2023).

33. Safe Superintelligence Inc. <https://ssi.inc>.
34. Nick, B. Superintelligence: Paths, dangers, strategies. (2014).
35. A Narrow Path. <https://www.narrowpath.co/phase-0>.
36. Bengio, Y. *et al.* Managing extreme AI risks amid rapid progress. *Science* **384**, 842–845 (2024).
37. Clymer, J., Gabrieli, N., Krueger, D. & Larsen, T. Safety Cases: How to Justify the Safety of Advanced AI Systems. Preprint at <https://doi.org/10.48550/arXiv.2403.10462> (2024).